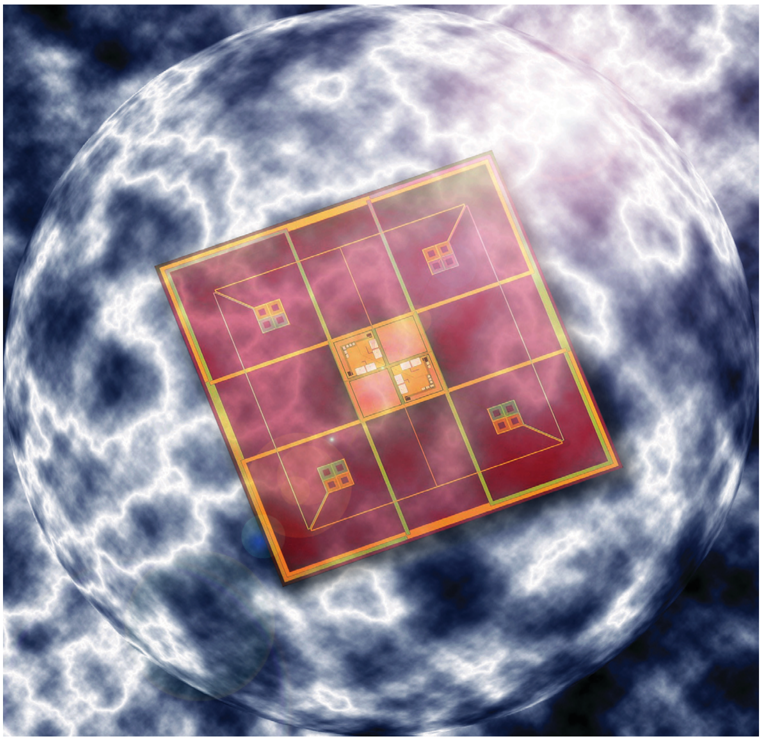
Elekta Neuromag

Internal active shielding

User's Manual



ELEKTA

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Contents

List of symbols

The following symbols are used in the system and in the manuals. **Familiarize yourself with each symbol and its meaning before operating this system.**

Caution, consult accompanying documents. Parts of the system are marked with this symbol when it is necessary for the user to refer to important operating and maintenance instructions given in the manuals accompanying the system. In the manuals, it also calls attention to specific instructions. These instructions may contain procedures, practices, conditions or the like which must be correctly performed or adhered to in order to ensure safe operation and to avoid direct damage to the patient, operator, or the system.

Consult instructions for use. Parts of the system are marked with this symbol when it is necessary for the user to refer to important operating and maintenance instructions given in the manuals accompanying the system. In the manuals, it also calls attention to specific instructions. These instructions may contain procedures, practices, conditions or the like which must be correctly performed or adhered to in order to ensure correct operation and/or increased safety and to avoid damage to the system.

Type BF (body floating) equipment symbol. The applied parts (parts in direct contact with the person being investigated with the system) and the type plate are marked with this symbol to indicate that they fulfill the leakage current requirements of the safety standard IEC 60601-1.

Ground (earth) terminal symbol. Used to identify terminals which are intended for connection to an external ground conductor for functional reasons.

Protective ground (earth) terminal symbol. Used to identify terminals which are intended for connection to an external ground conductor for protection against electrical shock.

Static electricity symbol. The parts of the system marked with this symbol indicate the presence of components susceptible to static electricity and require the use of special static-electricity preventing techniques. See *Elekta Neuromag System Hardware: User's Manual*.

Non-ionizing radiation, RF transmitter. Marking on equipment or equipment parts that include RF transmitters or that intentionally apply RF electromagnetic energy.

Separate collection of waste electrical and electronics equipment (WEEE) necessary (European Union directive 2002/96/EC on WEEE).

Date of manufacture: year (four digits) followed by month.

1

1 General

1.1 Brief description of the product

The internal active shielding system is a unique magnetic shielding technique intended to optionally provide additional magnetic shielding for the Elekta Neuromag® magnetoencephalograph.

The active shielding system increases the dynamic range of the magnetometers, related to the external magnetic interferences, by internal feedback compensation that uses the sensor array of the biomagnetometer as a zero indicator and compensation coils placed inside the magnetically shielded room to deliver a cancellation field for attenuating the interference.

1.2 Intented use of the product

The internal active shielding system is intended for shielding a biomagnetometer (MEG) from external magnetic disturbances. It is intended to be an optional component of the Elekta Neuromag® magnetoencephalograph.

1.3 How to use this manual

This manual describes only the principles and use of the internal active shielding system. For instructions on using the Elekta Neuromag® MEG system, please refer to the following hardware and software manuals delivered with the system:

NM20215A-*	Elekta Neuromag® System Hardware User's Manual
NM20216A-*	Elekta Neuromag® System Hardware Technical Manual
NM10238A-*	Data Plotting User's Guide
NM10239A-*	Signal Processor User's Guide
NM20568A-*	Source Modelling Software Users Guide
NM21283A-*	Sensor Tuner User's Guide
NM23065A-*	Data Acquisition User's Manual

NM21993A-*	MaxFilter™ 2.x User's Guide (where available)
NM21492A-*	MaxFilter™ 1.x User's Guide

2

2 Safety information

2.1 Warnings

CAUTION! This section contains important information concerning the safe use of the system and maintaining reliable operation. Read the instructions entirely before using the system.

2.1.1 Magnetic materials and magnetic fields

Magnetic materials inside the shielded room may distort the field inside, degrading the performance of the active compensation system. Strong magnetic fields will also magnetize the magnetically shielded room wall material to saturation, degrading the passive shielding performance.

NOTE! Do not bring strong permanent or electric magnets inside or even close to the magnetically shielded room. Do not attach the magnet to the magnetically shielded room wall, floor, or ceiling. Avoid bringing magnetic material inside the room, especially during measurements.

Strong magnetic fields in the immediate vicinity of the sensors may also cause magnetic flux to be trapped in the superconducting thin films. Trapped magnetic flux in the dc SQUID manifests itself usually as deteriorated signal quality due to increased noise or disturbances. Normal performance can, however, be recovered by detrapping. For futher information on how to avoid trapping and how to recover from flux trapping refer to *Elekta Neuromag® System Hardware User's Manual*.

For proper operation of the internal active shielding system, it is important that also electrically conducting materials with large area (> 0,25 m^2) are not brought near the inside walls, floor, or ceiling.

NOTE! Avoid large electrically conducting loops near inside walls, floor, or ceiling.

The internal active shielding system has been tuned during the installation of the Elekta Neuromag system. The tuning parameters are not directly accessible by the user.

NOTE! Do not adjust the parameters of the active shielding system.

To achieve optimal active shielding performance and signal quality, the door of the magnetically shielded room should be closed properly when making a measurement. Refer to instructions given in Section 4.

NOTE! Make sure that the door of the magnetically shielded room is closed properly before starting a patient measurement.

2.1.2 Grounding and electrical safety

The grounding system of Elekta Neuromag® has been carefully designed and realized. Do not add any equipment to the system or change any cabling without considering the possible side-effects. If in any doubt, contact Elekta.

The grounding system is based on single-point grounding. This is very important for the proper functioning of the active shielding system since otherwise ground loops will be formed, resulting in artefacts in the measurements. On the other hand, to maintain electrical safety, the system must be grounded at all times. Especially, the walls, ceiling, and floor of the magnetically shielded room will be electrically grounded.

All interior surfaces of the room will be covered by electrically insulating inner decoration to avoid accidental grounding of the patient as required by the medical device safety standards. However, grounding may be compromised by conductivity formed by eventual liquids between the shielded room structures and other parts of the building.

NOTE! The system must not be grounded to any other place than the main grounding point. The magnetically shielded room shall be otherwise electrically isolated from all building parts, including ventilation ducts, helium exhaust pipes, cables, etc. Avoid any fluids from penetrating into the room structures, both inside and outside.

CAUTION! The grounding cables must not be disconnected. Do not remove the inner decoration or cable covers.

Leave maintenance of the active shielding system to trained service personnel. There are no operator serviceable parts.

2.1.3 Other precautions

The internal active shielding system is designed to be used together with the MaxFilter™ interference elimination software. When internal active shielding is used, MaxFilter™ must be applied on the recorded data before it may be analyzed.

The application of MaxFilter™ will provide optimal interference suppression and spatial representation of the signals. Before using the Elekta Neuromag® system with active shielding, the users must be trained to operate the system.

CAUTION! Before analyzing data recorded with internal active shielding, MaxFilter™ must be applied on the data according to the instructions provided in *MaxFilter™ User's Guide*. Train all users before using the system.

CAUTION! The internal active shielding system is only intended to be used with the Elekta Neuromag® system and its accessories.

2.2 EMC

NOTE! Medical electrical equipment needs special precautions regarding electromagnetic compatibility (EMC). It needs to be installed and put into service according to the EMC information provided in this manual and *Elekta Neuromag® System Hardware Technical Manual.*

NOTE! Portable and mobile radiofrequency (RF) communications equipment can affect medical electrical equipment.

NOTE! Radiofrequency (RF) intereference may deteriorate signal quality and lead to incorrect results.

To maintain high level of electromagnetic interference immunity, some precautions are necessary. The internal active shielding system is intended to be operated with the door of the magnetically shielded room properly closed during measurements with the Elekta Neuromag® system. Furthermore, all cables entering the magnetically shielded room must be properly filtered, and radiofrequency transmitters, such as mobile phones, mains operated devices, as well as active digital electronics inside the magnetically shielded room, must be avoided alltogether.

Use of the stimulator cabinet outside of the magnetically shielded room is highly recommended for all other equipment. The stimulator cabinet forms part of the RF enclosure, and there is a direct access from the cabinet to the inside of the shielded room. The cabinet is equipped with signal and mains feedthrough filters. Place, for example, the isolation units of somatosensory stimulators inside the stimulator cabinet.

To avoid radiated intereference via cabling, digital devices that are active during measurement should, however, be avoided also inside the stimulator cabinet, and only third-party devices compatible with the system should be used.

NOTE! All cables running from outside to inside of the magnetically shielded room must be properly filtered or contained in the stimulator cabinet that forms part of the RF enclosure. Do not use mobile phones inside the magnetically shielded room.

3

3 Product description

The internal active shielding system employs the sensor array of the probe unit to measure the residual ambient field variations inside the magnetically shielded room. These signals are fed back to the coils inside the magnetically shielded room, forming a closed control loop that effectively minimizes the external disturbances at the sensor area.

The six coils inside the room can be individually controlled. Therefore, in addition to the homogeneous fields B_x, B_y, and B_z it is also possible to compensate uniform gradients $\partial B_x/\partial x$, $\partial B_y/\partial y$, and $\partial B_z/\partial z$ as illustrated in figure 3.1. However, there are only two independent diagonal uniform gradient components since, according to Maxwell's equations, the sum of the diagonal gradients

$$\partial B_x/\partial x + \partial B_y/\partial y + \partial B_z/\partial z = 0.$$

There are thus five independent control loops, three homogeneous field components and two diagonal gradients, in the internal active shielding system. These will be called *virtual compensation channels*.

The hardware for the internal feedback compensation system is shown in figures 3.2 and 3.3. The currents to drive the feedback coils are connected with an electronics cable plugged into a con-

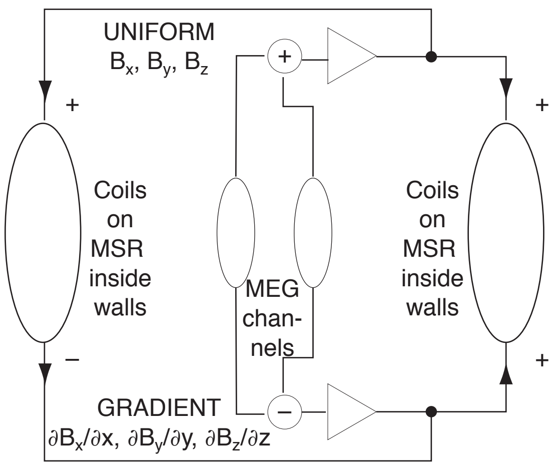

Figure 3.1 Priniciple of the internal active shielding

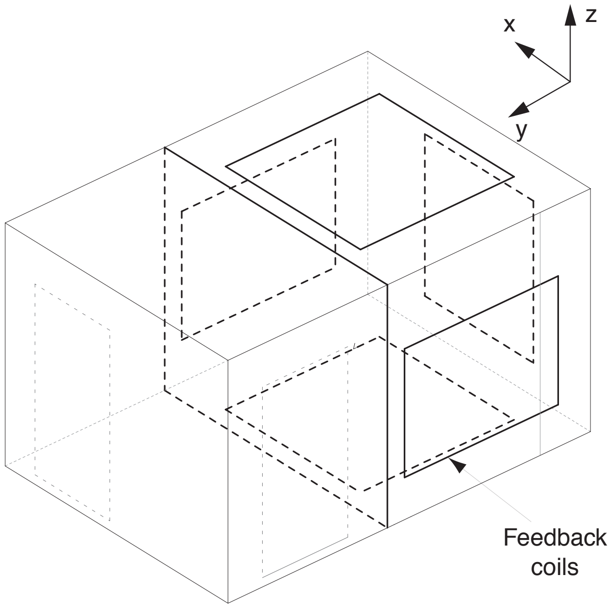

Figure 3.2 The internal active shielding coils inside the magnetically shielded room.

nector (SCSI II type) on a junction box cover in the back wall of the magnetically shielded room.

Each virtual channel receives input from a set of actual MEG channels as a weighted linear combination of the MEG outputs. The signals are transmitted over the system backplane bus to a feedback controller board which calculates the feedback signals

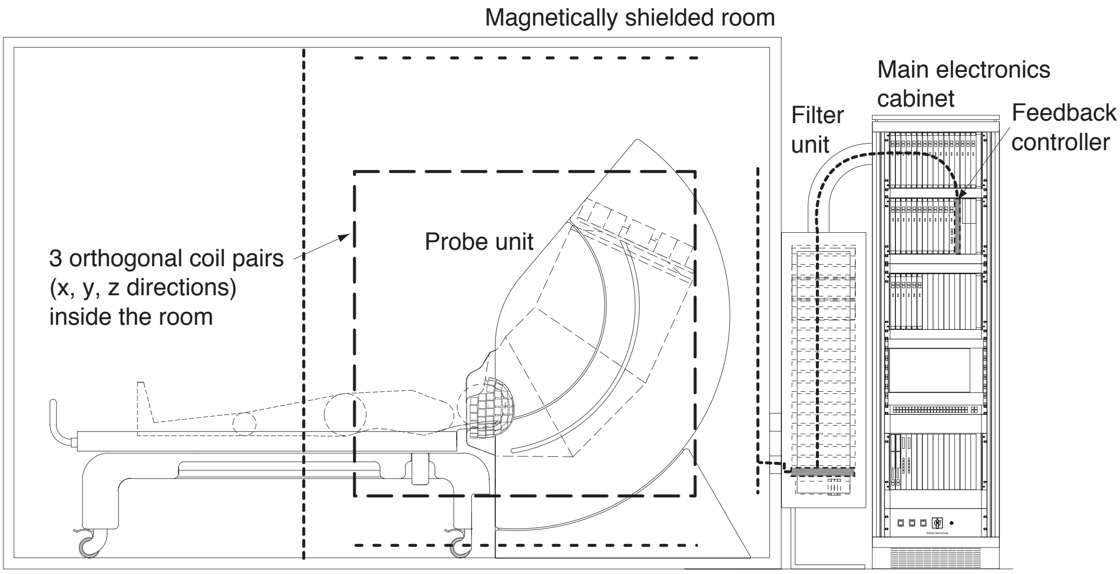

Figure 3.3 The hardware components of the internal active shielding system

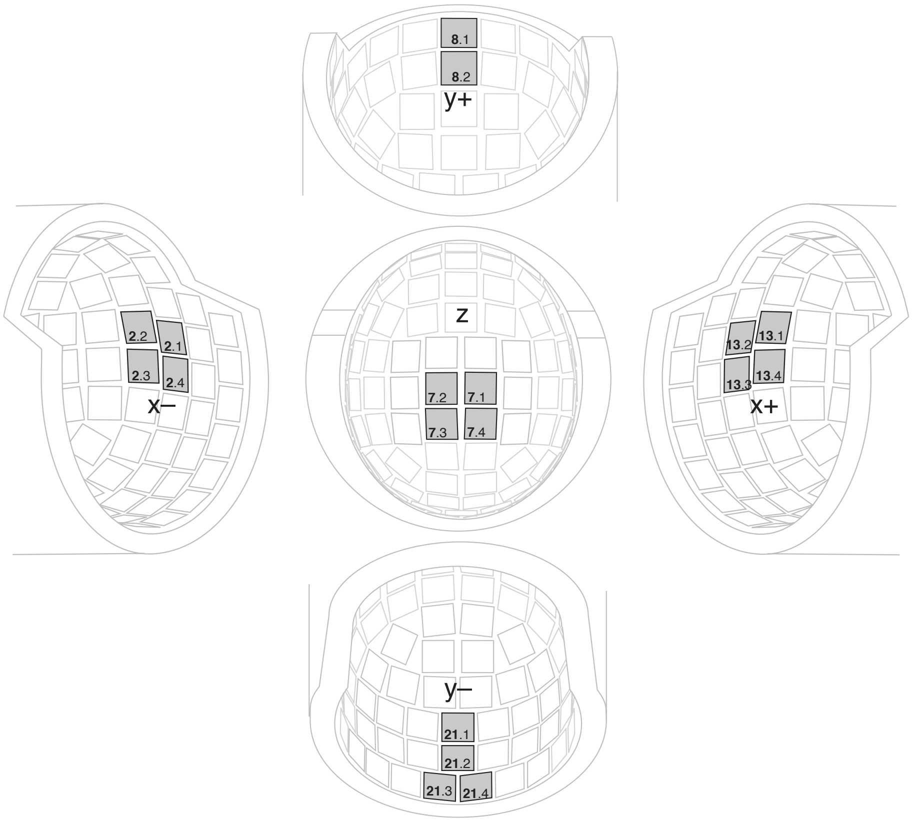

Figure 3.4 Virtual compensation channel inputs (magnetometer channels only)

by using a proportional-plus-integrating control algorithm for each virtual compensation channel. The control algorithm tries to keep the virtual compensation channels representing the uniform field and diagonal gradient components zero by generating a feedback that cancels any changes due to external fields. The feedback controller finally calculates the feedback coil drive currents and distributes them to the appropriate coils.

The five virtual compensation channel signals representing feedback necessary to cancel the uniform field and diagonal gradient components are also sent into the normal data stream to the data acquisition workstation as channels MISC201–MISC206.

These channels refer to the standard *head coordinate system,* where the x-axis points from the midpoint of the head towards the right ear, the y-axis from the head midpoint towards the nose, and the z-axis towards the vertex according to table 1.

Table 1. Virtual compensation channels (head coordinates)

MISC201:	B_x	MISC202:	$\partial B_x/\partial x$
MISC203:	B_y	MISC204:	$\partial B_y/\partial y$

MISC205:	B_z	MISC206:	N/A

Also the six real coil drive signals are sent into the normal data stream to the data acquisition workstation as channels MISC301–MISC306. Note that they refer to the *room coordinate system* (see Figure 3.2), where the x-axis is directed along the transversal (short) axis of the room, the y-axis along the longitudinal (long) axis of the room, and the z-axis upwards.

The actual mapping between the head and room coordinates depends, of course, on the measurement position of the probe unit. Distribution of signals to the coils will be different in supine and seated measurement positions. Before measurement, the gantry position parameter must be set according to instructions given in Section 4. The coil currents are assigned according to table 2.

Table 2. Coil current channels (room coordinates)

MISC301:	*x*–	MISC302:	*x*+
MISC303:	*y*–	MISC304:	*y*+
MISC305:	*z*–	MISC306:	*z*+

3.1 MaxFilter™

The standard MEG channel sets used for the active compensation are shown in figure 3.4. As normal magnetometer channels are used for the compensation system, part of the MEG signals can contribute to the feedback currents. This is, however, easily taken into account by applying MaxFilter™ after data acquisition. The filtering then restores the original field pattern without the effect of external interference. Refer to *MaxFilter™ User's Guide* for further instructions.

To prevent accidental analysis of non-filtered data, the data files are marked as being recorded with internal feedback, and the data analysis programs will not allow such files to be used before MaxFilter™ has been applied.

NOTE! The Signal Processor (version 3.4 or above) allows non-filtered data to be reviewed before application of MaxFilter™. This functionality is provided exclusively for troubleshooting purposes.

The MaxFilter™ software separates field components originating from inside and outside the detector array. External interference and feedback coil signals fall into the outside components whereas actual brain signals fall within the inside components.

For further details, refer to the *MaxFilter™ User's Guide*.

4

4 Operating instructions

This manual will only give guidance on issues specific to the internal active shielding system. Instructions on operating the Elekta Neuromag® MEG system hardware and software are given in the reference manuals listed in section 1.3.

4.1 Activating internal active shielding

To activate the internal feedback active compensation, perform the following steps:

1. Start the Elekta Neuromag® Data *Acquisition* program.

   Refer to *Elekta Neuromag® System Hardware User's Manual* and *Data Acquisition User's Manual* for further details on how to prepare and set up a measurement.

2. In the main *Acquisition* window (figure 4.1), set the gantry position to correspond to the physical position by pressing the lowermost *Change* button to toggle between supine and seated positions.

NOTE! Before starting a measurement, set up the correct gantry position parameter first.

   As described in Section 3, it is important to set the gantry position parameter first to correspond to the physical position. It will set the proper active feedback parameters and the proper signal space projection (SSP) vectors needed for on-line software interference rejection (see *Data Acquisition User's Manual*).

3. In the main acquisition window (figure 4.1), open the *Acquisition parameters* dialog box by pressing the Change acquisition button (third button from the top).

   Note that the status of internal active shielding is shown on the text line to the right of the button.

4. In the *Acquisition parameters* dialog box (figure 4.2), select the *Use MaxShield* button to activate internal active shielding.

NOTE! The operation frequency of the head position indicator coils is set to be above the bandwidth within which active feedback is effective. As a result, the sampling frequency cannot be selected arbitrarily low. In practice, the minimum sampling frequency is limited to 1,000 Hz. The acquisition will not start if a value lower than this is selected.

4.2 Monitoring data acquisition

For monitoring the operation of the internal active shielding system, the data acquisition software provides two predefined channel selections for the raw data display of the data acquisition software associated with the active feedback.

Perform the following steps to activate either of the predefined channel selections:

1. In the Raw data display, click *Selection...* to open the *Channel selections* window.
2. In the *Channel selections* window, select either *MaxShield* button or the *MaxAll* button.

   The *MaxShield* channel selection (figure 4.3) displays the virtual feedback channels MISC201–MISC206 and the coil currents MISC301–MISC306 (see Section 3 for further explanation).

   The *MaxAll* channel selection (figure 4.4) displays the above virtual feedback channels and coil currents as well as the actual MEG channels used for formation of the virtual feedback channels.

   The MISC201–MISC206 and MISC301–MISC306 signals are displayed in voltage units, and the display scale can be set from the *Scales* dialog box of the raw data display (parameter “MISC scale”). The signal voltage corresponds to control voltage to drive the coils, and 1 V corresponds approximately to 1 mA. The signals show usually slow variations following external disturbances. Note that MISC206 will show a straight line (see explanation in Section 3).
3. If the data displayed shows instabilities, reset all channels by selecting *Tools* > *Reset channels*.

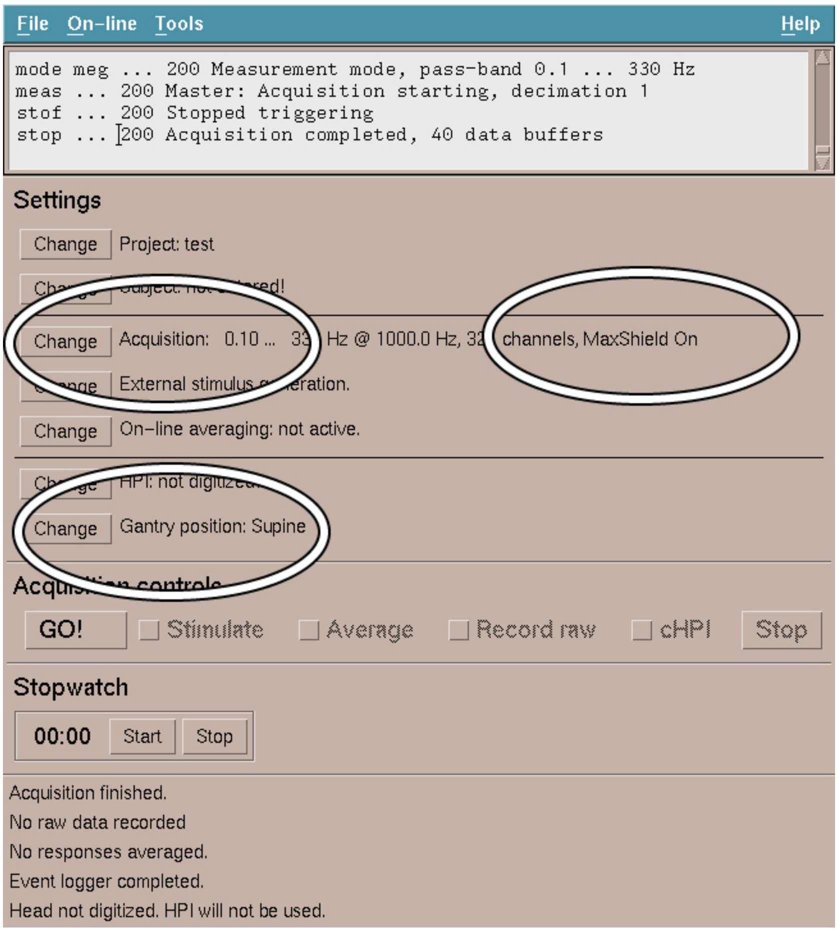

Figure 4.1 The main acquisition window.

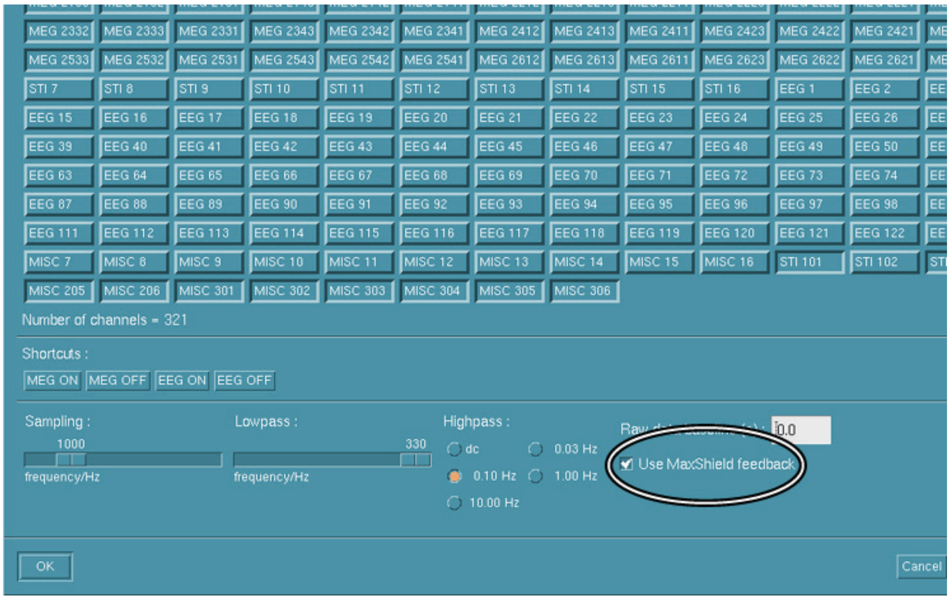

Figure 4.2 Close-up of the acquisition parameters dialog.

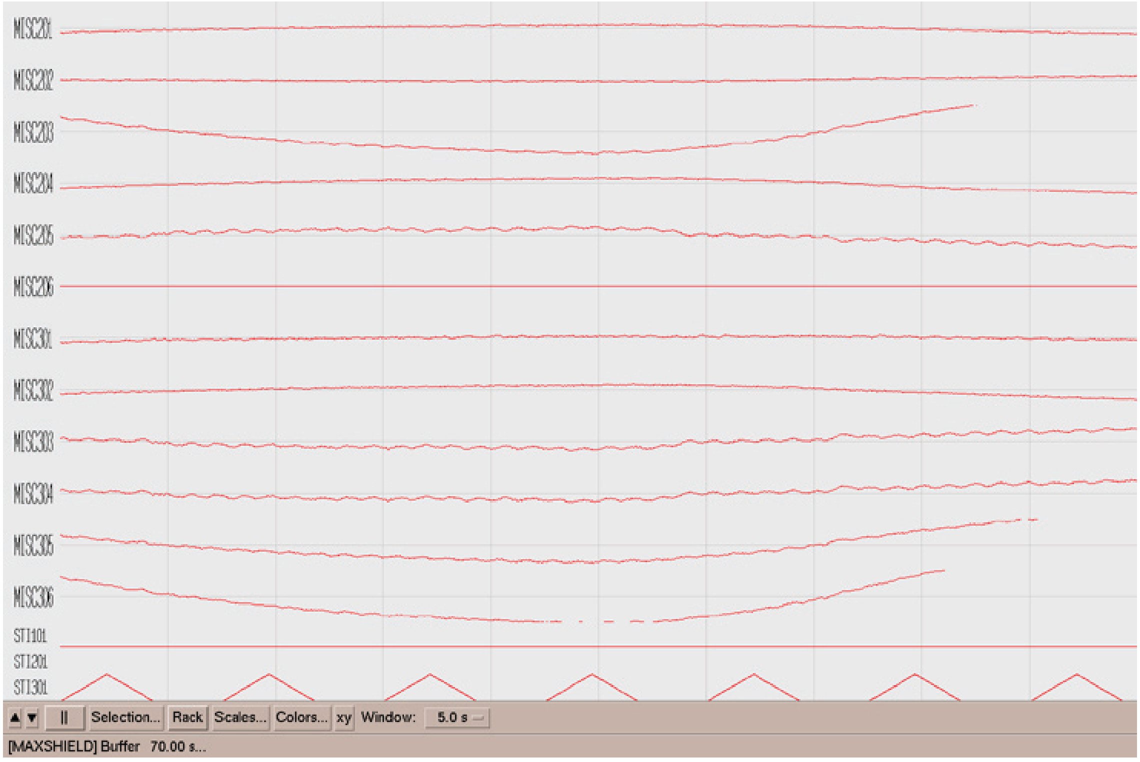

Figure 4.3 The “MaxShield” channel selection in raw data display.

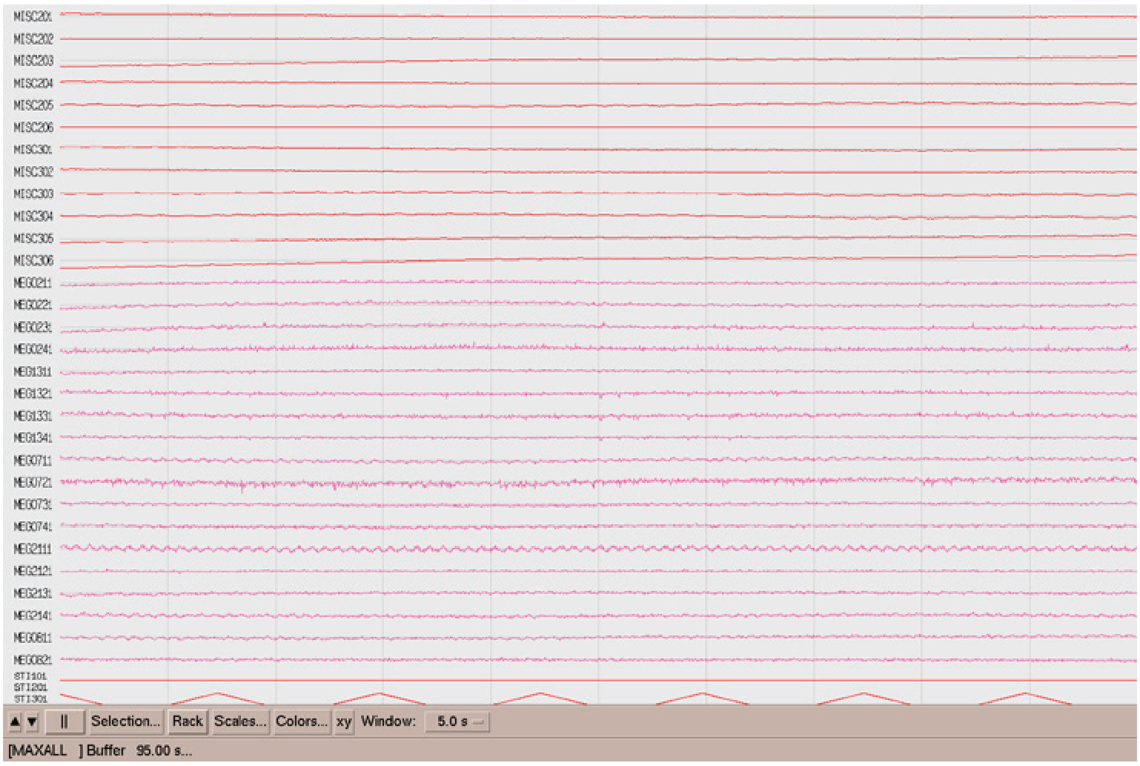

Figure 4.4 The “MaxAll” channel selection in raw data display.

4.3 Deactivating internal active shielding

To deactivate the internal feedback active compensation, perform the following steps:

1. Start the Elekta Neuromag® *Acquisition* program.
2. In the main acquisition window (figure 4.1), open the *Acquisition parameters* dialog box by pressing the Change acquisition button (third button from the top).

   Note that the status of internal active shielding is shown on the text line to the right of the button.
3. In the *Acquisition parameters* dialog box (figure 4.2), deselect the *Use MaxShied* button to activate internal active shielding.

4.4 Tuning

To tune the sensors of the system, perform the following steps:

1. Deactivate internal active shiedling as shown in section 4.4.

   During tuning, the internal active shielding must be switched off.
2. Tune the sensors according to the instructions provided in *Sensor Tuner User's Guide*.
3. Activate internal active shielding as shown in section 4.2.

4.5 Analyzing data

Perform the following steps to analyze data acquired with internal active shielding:

1. Apply MaxFilter™ on the data as described in the *MaxFilter™ User's Guide*.

NOTE! Data recorded with the active feedback on must be run through MaxFilter™ before further analysis.

   Notice that the data may be previewed before applying MaxFilter using Signal Processor (Elekta Neuromag® Data Analysis Software version 3.4 and above). However, no analysis may be performed on the unfiltered data.

2. Analyze the data using any of the analysis tools provided with Elekta Neuromag®.

5 Maintenance

5.1 Checking of system performance

For checking system performance and data quality, follow the guidelines given in *Elekta Neuromag® System Hardware User's Manual*. For inspecting the operation of the feedback system the raw data display sets "MaxShield" and "MaxAll" explained in Section 4 may prove to be useful. In normal operation the signals show slow variations following external interference.

5.2 Troubleshooting guide

To aid troubleshooting, some common causes of trouble are listed below.

Very large external interference on signals

- Check that the magnetically shielded room door has been properly closed.
- Check that the gantry position parameter has been correctly set.
- Ascertain that the internal active shielding system is active by checking the main window status message and by monitoring the virtual feedback channel displays (MaxShield or MaxAll).
- Check that the virtual channels are not saturated. You may need to adjust the "MISC" scale and to toggle the "Remove Dc offset" selection.
- Check that the internal active shielding cable from the electronics rack to the inside coils at the back wall of the magnetically shielded room is connected.
- Check that the SSP vectors are on from the "Scales" display of the raw data display.
- Try resetting the channels two-three times if necessary (allow some time for the channels to settle).
- Check if there are unusually strong interference sources in the vicinity of the magnetically shielded room.

Noisy channels

- Try resetting the channels two-three times if necessary (allow some time for the channels to settle).
- Check that the gantry position parameter has been correctly set.
- Tune the channels. Proper tuning of channels belonging to the set from which the virtual feedback signals are formed is especially important.
- Check that the virtual channels are not saturated. You may need to adjust the “MISC” scale and to toggle the “Remove Dc offset” selection.
- Check that the internal active shielding cable from electronics to the inside coils at the back wall of the magnetically shielded room is connected.

Channels missing from acquisition

- Check the indicator lights of the electronics boards of the main electronics cabinet and reset the corresponding unit if necessary. Refer to *Elekta Neuromag® System Hardware User’s Manual* for further instructions.
- Check that the optic fibre running from the main electronics cabinet to the acquisition workstation is not damaged and not bent to a very small radius.

5.3 Maintenance

There is no dedicated maintenance program for the internal active shielding system. Eventual service operations are performed in conjunction with maintenance of the Elekta Neuromag® system and includes checking of the operation of the active shielding system and retuning it if necessary.

All service operations must be left to service personnel trained and qualified by Elekta.

Circuit diagrams, component part lists, descriptions, calibration instructions, or other information assisting in maintenance will be available for service personnel on request.

5.4 Spare parts

Spare parts for the active compensation system are available as normal Elekta Neuromag® spare parts. Consult your Elekta service representative for more information.

6 Technical data

6.1 Active shielding

- Six internal feedback compensation coils mounted on the interior walls of the enclosure (individually driven)
- Feedback compensation coil currents max ±10 mA
- Power requirement for the feedback compensation: included in the Elekta Neuromag® MEG system

6.2 Shielding factor

Frequency	Shielding factor
0.01 Hz	20 dB
0.1 Hz	20 dB
1 Hz	20 dB
10 Hz	10 dB
100 Hz	6 dB

The shielding factors exclude the contribution of passive shielding. Typical measurement uncertainty is about 6 dB.

6.3 Environmental specifications

- Maximum magnetic fluctuations within the magnetically shielded room: 50 nT peak-to-peak (during operation)
- Maximum remanent DC field within the magnetically shielded room: 50 nT

A

A Appendix

Basic concepts, terminology

Applied part

Parts of the system in direct contact with the person being investigated with the system.

BF type

Body floating. An applied part providing enhanced protection against electrical shock. An applied part of BF type must fulfill leakage current requirements specified in the standard IEC 60601-1 even in the case that the patient is unintentionally connected to an external mains voltage.

Class I device

A device whose protection against electrical shock does not rely only on basic insulation but also has a permanently connected protective earth connection so that accessible metal parts cannot become live if basic insulation is damaged.

Feedback

A method where the output of a sensor is amplified and fed back to the sensor input. In this configuration, the sensor acts as a null-detector while the feedback effectively compensates for the input from the outside. The feedback signal is thus equal to the external input.

Flux transformer

A superconducting circuit comprising at least two coils connected in series. Because of superconductivity, magnetic flux is conserved. Thus, a magnetic field imposed on one of the coils will induce a shielding current to flow in the circuit, cancelling the effect of externally imposed flux. Since the shielding current will induce a magnetic field in other coils belonging to the circuit, a flux transformer can be used to scale magnetic field intensity by varying the coil parameters.

Gradiometer

A flux transformer coupling external magnetic signal to the SQUID detector, making the SQUID to respond to spatial variations of the external magnetic field. A gradiometer comprises a multiple pickup coil and a signal coil which couples the signal to the SQUID. The gradiometer is insensitive to uniform magnetic fields.

IEC60601

A family of international standards concerning the safety of medical electrical equipment and systems.

Leakage current

Current that is not functional. For example, patient leakage current is the unintentional current that flows from equipment via patient to ground.

Magnetically shielded room (MSR)

A special enclosure whose walls, floor, and ceiling are made of plates of high-permeability alloy (μ-metal) and of high-conductivity metal (typically aluminium). The room distorts the external magnetic field in such a way that the magnetic field inside is substantially weaker. The shielding efficacy increases with frequency.

Magnetometer

A flux transformer coupling external magnetic signal to the SQUID detector, making the SQUID to respond to the external magnetic field. A magnetometer comprises a single pickup coil and a signal coil coupling the signal to the SQUID.

SQUID

Superconducting QUantum Interference Device: The SQUID is an ultrasensitive magnetic flux detector based on superconductivity and so-called Josephson effect. It operates at cryogenic temperatures.

Superconductivity

A state where electric resistance equals zero. Present in several substances, e.g. in Niobium and Lead. A superconductor also repels magnetic flux, i.e., magnetic field lines cannot penetrate the superconductor.

Trapped flux

A phenomenom where magnetic flux is trapped in supreconducting structures, e.g., thin films. A superconductor normally repels magnetic flux. However, if regions of the superconductor become non-superconducting as a result of being exposed to strong magnetic fields, magnetic field lines can enter the superconductor and become trapped inside superconducting areas. May lead to deteriorated operation and increased noise of the SQUID sensor. Movements of the trapped flux are seen as magnetic signal jumps. Trapped flux can be released by heating the superconductor above its superconducting transition temperature, typically over 10 K.

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